

8/10

Biochemical Engineering Project Due on April 10

Design the experiments to express the following protein at the production scale. When you select an expression system, you need to consider the cost, yield, and feasibility (size, glycosylation and disulfide bond) of using the selected expression system. Your report should include

1. DNA sequence of the protein:
 - a. and other component for expression in the sequence
2. Molecular cloning procedures
 - a. describe the key steps for the cloning, selection, and verification (gene and protein)
 - b. provide the detailed information about expression vector and cloning site

MNLQAQPKAQNKRRCLFGGQEPAPKEQPPPLQPPQQSIRVKEEQYLGHEGPGGAVSTS
QPVELPPPSSLALLNSVVGPER TSAAMLSQQVASVKWPNSVMAPGRGPERGGGGGVS
DSSWQQQPGQPPPHSTWNCHSLSLYSATKGSHPG VGVPTYYNHPEALKREKAGGPQL
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PPPPNPVAAFPPQKQQQQQQQPPQQQQQQQAALPQMPLFENFY SMPQQPSQQPQDFGLQ
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EGILPPSALDGAGTQPGQEATGNLFLHHWPLQQPPGSLGQPHPEALGFPLELRESQLLP
DGERLAPNGREREAPAMGSEEGMRAVSTGDCGQVLRGGVIQSTRRRRRRASQEANLLTL
AQKAVELASLQNAKD GSGSEK RKSVLASTTKCGVEFSEPSLATKRAREDSGMVPLIIPV
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frequency table):

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 aaaaaaaaaaacggaaccttttagcaaaaccagaaccaggaaaacaccttccgtgc
 aaaaatgcggccgcgaattc

Red part is Eco RI cutting site.

According to NCBI's database, the putative functions are:

1. ELM2 super family. The domain is usually found to the N-terminus of a myb-like DNA binding domain pfam00249. ELM2 is also found associated with an ARID DNA binding domain. Hence this protein may be involved in DNA binding.
2. 'SWI3, ADA2, N-CoR and TFIIB' DNA-binding domains. Tandem copies of the domain binds to telomeric DNA repeats as part of capping complex. This domain also found in regulatory transcriptional repressor complex which also binds to DNA.

Detail Information:

Basic Strategy: The pET system for protein expression requires IPTG which is a very expensive inducer. This feature makes pET system not fit for large scale protein production. The pBAD system use L-arabinose as inducer and glucose as repressor for protein expression. L-arabinose and glucose are both very cheap substances which makes pBAD suitable for large scale production. And pBAD system could precisely control the expression of protein give manufacturing convenience.

Expression Host: *E. coli* strain contain genotype araBAD⁻ or araEFGH⁺. Invitrogen provided TOP10 and LMG194. Strain BW25133, BL21 and BW26434 could also be used as host cell.

Expression System: pBAD.

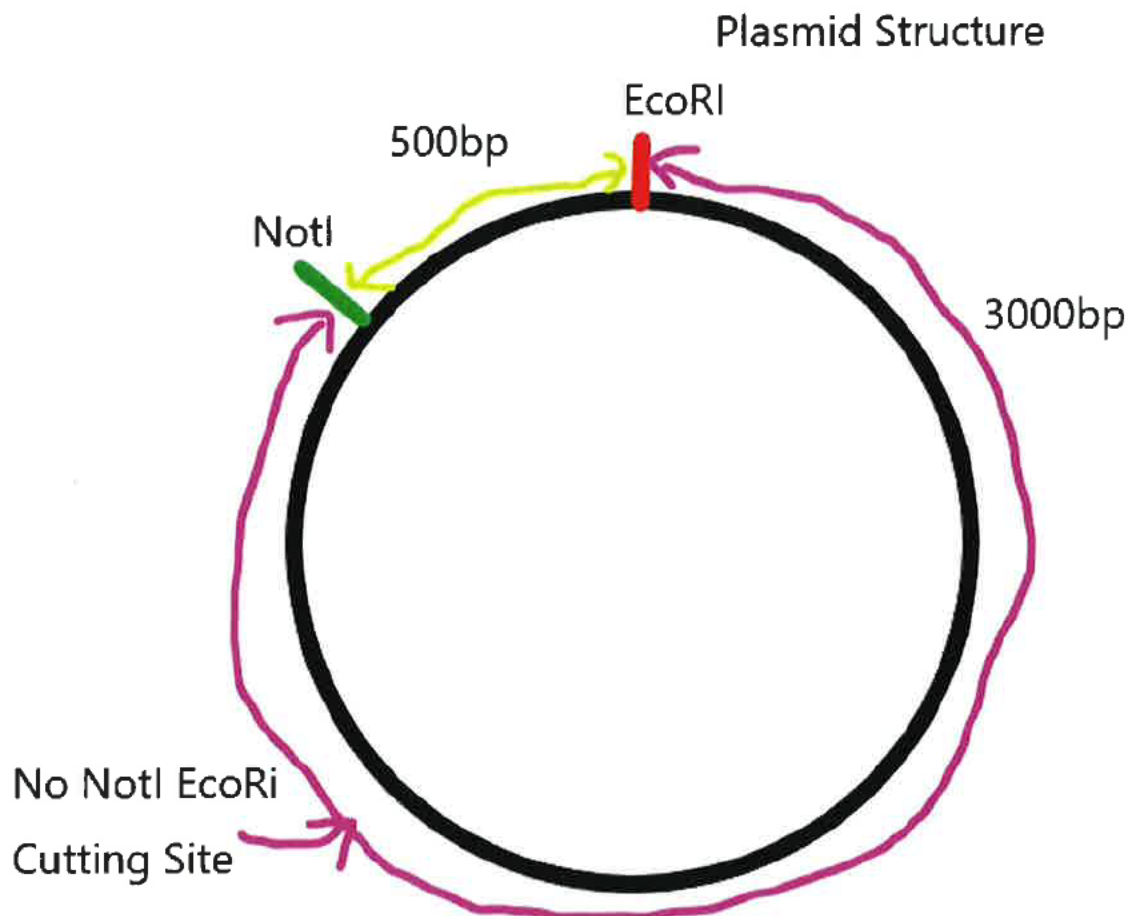
Sequencing Primer:

1. CAGCCGAAAGCGCAGAACAAAC
2. AAGAACAGTATCTGGGCCATGAAG
3. AGCGTGAAATGGCCGAACAG
4. GCATAGCACCTGGAAGTGCATAG

Expression Condition:

0.1%~0.2% L-arabinose, chaperone, 18 celcius degree, time(to be determined)

Vector Structure(only indicating enzyme cutting site within MCS) :



Vector also contains ampicillin resistant gene, araC, pBAD, pBR ori.

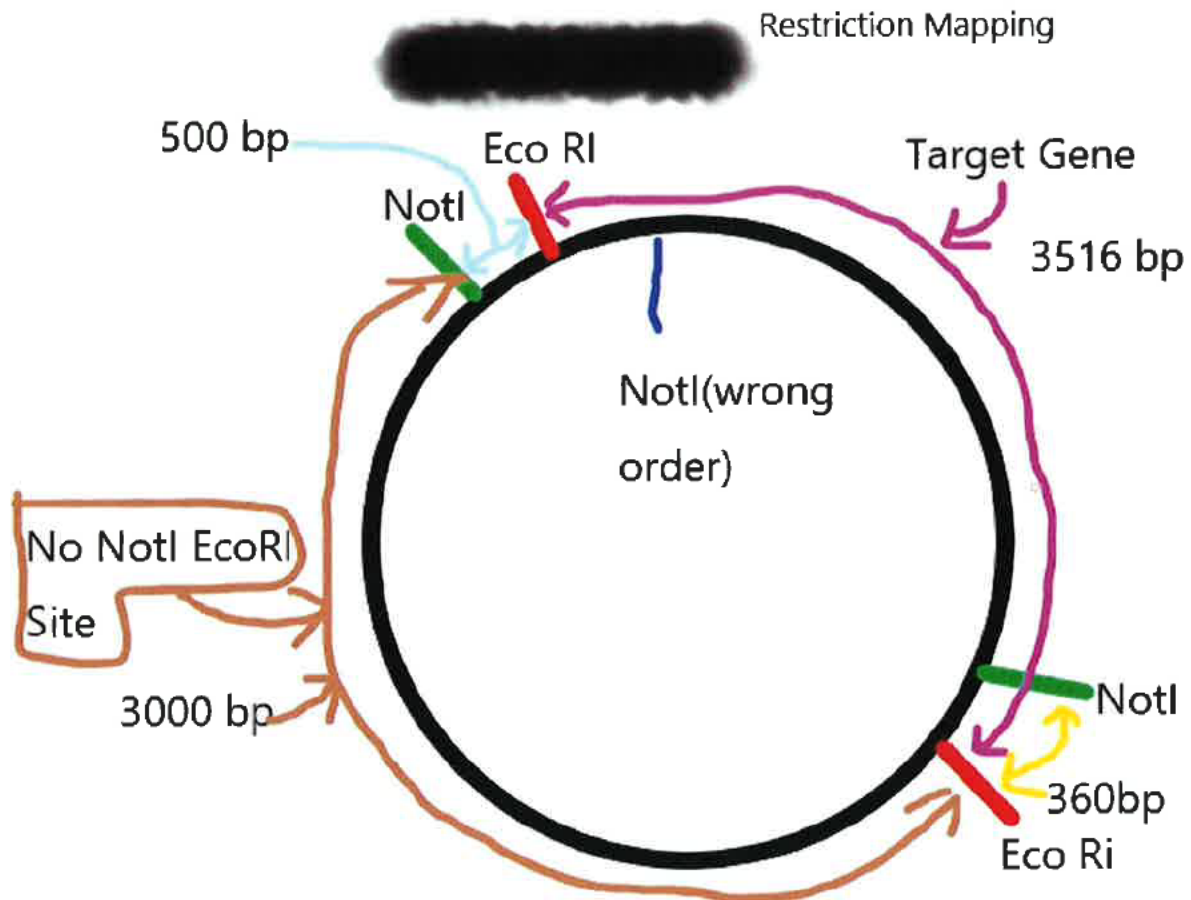
Detailed Vector Structure:

According to <http://www.restrictionmapper.org/>, this target gene does not have Eco RI cutting site.

And has one NotI cutting site at position 3156. Hence plasmid structure is as above.

Detailed Steps:

1. Convert protein into codons. Purchase gene from company. Amplify gene using PCR method.
PCR primer (cloning primer): http://bioinfo.ut.ee/cgi-bin/primer3-0.4.0/primer3_results.cgi
Left primer: attagcaccagcaccattcc
Right primer: ccacaccagatccgctttat
2. Restriction endonuclease cut (Eco RI) both vector and gene. T4 ligase is used to form loaded vector.
3. Use another endonuclease (NotI) cut loaded vector to obtain restriction map to make sure gene is inserted in a correct order. Shown in picture below.



After cutting with NotI, run an electrophoresis. If result indicating two bands of close bp numbers (3360 bp vs 3656 bp), then target gene is loaded correctly. If otherwise (860 bp vs 6236 bp), then target gene is not correctly loaded.

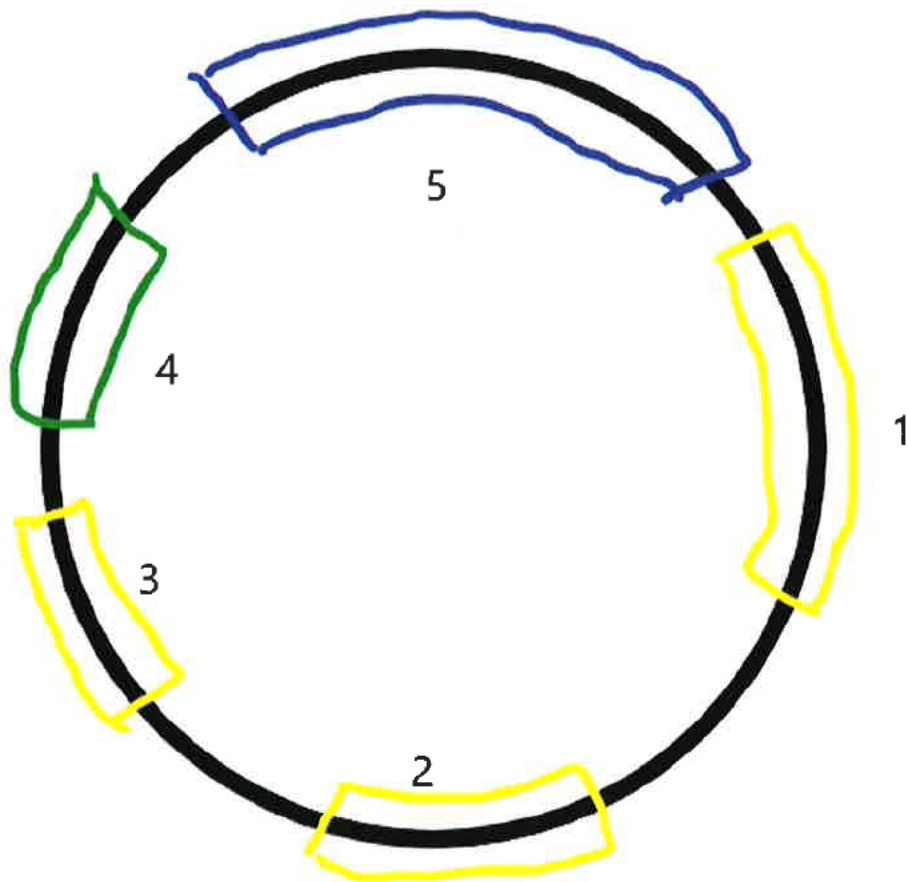
4. Transform correctly loaded vector into transparent E. Coli cell. Use ampicillin as selectiver pressure to force E Coli cell to uptake vector. Use ampicillin and kanamycin to select which cell is loaded with vector. If cell can grow in dish containing ampicillin but can not grow in dish containing kanamycin, that cell has correct loaded vector.

5. Cell growth step. During this step, we need to suppress protein expression by using glucose. Let cell grow to a certain number and then start protein expression. Cell grow in an environment containing glucose, ampicillin and ATP.
6. Remove glucose and add L-arabinose as inducer to trigger expression of protein. Protein is expressed at 18 C to reduce probability of misfolding. Also chaperon is used to help protein to fold correctly. But we need to find out if 18 C will slow down protein expression. Solubility is another important issue. Inclusion bodies could be helpful and reduce purification cost. Find balance and optimization between misfolding rate, precipitation rate and expression rate. All of these should start from lab level experiment then scale up into manufacture.
7. When protein expression reached a certain extent, stop expression by adding glucose. Acquire crude eluent through lysis and then run mixture solution over a membrane contains Nickle beads. Nickle beads will grab six histidine group hence target protein is filtered out (with tags). Immobilized metalion affinity chromatography could also do the job.
8. His-tag protein will be cut with protease to obtain detagged protein. Through a subtractive IMAC pure protein is obtained. Run a gel electrophoresis (load cell extract, supernatant fraction, flow through fraction from IMAC, eluted protein) to confirm.
9. Protein sequencing. Edman does not fit. Run SDS-Page first. If result shows one obvious band, we could proceed to MALDI-MS for specific amino acid sequencing. If result shows more than one band then we need to separate all bands and do MALDI-MS to each band.



There are many parameters, conditions and environments need to be determined through large amount of experiments.

Detailed Vector Structure:



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1. Ampicillin resistance gene
 2. pBR322 Ori
 3. araC
 4. pBAD
 5. ATG, 6His, Xpress Epitope, Ek site, MCS. In MCS, two enzyme cutting site. One NotI cutting site and one Eco RI cutting site, as indicated in the picture 1.